

BUILT — RPS-safe stability suite (v0.63.0)

The data 'treasure vault' protects FILES; this adds a thin HEALTH layer that keeps the program running. All stdlib, no new deps, GPU never fatal — identical on modern / lite / legacy.

□ Preflight health gate (fail-fast)

- preflight.py checks python / free disk / DB quick_check / schema-vs-migrations BEFORE the server or OCR start.
- A real problem (disk full, corrupt index) STOPS with a clear message instead of crash-looping; schema drift -> WARN (run fix_schema_version).
- GPU is INFO only — absent on lite/legacy is fine, never a FAIL. Wired into run_ocr_auto.bat + run_app.bat.

Kills the crash-loop class of failure at the door.

□ Disk-space guard

- disk_ok() watches free space on the index drive (default 1 GB floor; env VIEWER_MIN_FREE_MB).
- OCR PAUSES a pass cleanly when low (the auto-runner retries); the page-render cache stops writing.
- Fail-OPEN: if free space can't be read, work continues — a probe glitch never halts the app.

The disk can't silently fill on a laptop.

□ Off-disk backup mirror

- safeguard.py mirror --to <USB/external/network> copies the vault to a SECOND location, verifying every file by SHA-256.
- One disk failure can no longer lose both the data and its backups. BACKUP-OFFDISK.bat wraps it.
- Daily snapshot+verify task already exists (register_snapshot_task.bat); this closes the single-disk gap.

Backups survive a dead drive.

□ Watchdog + stall detection

- watchdog_app.bat supervises the web server and AUTO-RESTARTS it if it ever crashes (kiosk/shop machine stays up).
- OCR now writes a heartbeat every batch; ocr_watchdog.py flags a hung pass (stale heartbeat) vs a healthy one.
- The auto-runner already restarts a pass that ENDS early; this catches one that HANGS.

Server stays up; a hung OCR pass is visible.

VERIFIED · RPS PARITY PRESERVED

preflight.py and ocr_watchdog.py compile + pass functional tests (gate goes no-go only on fatal checks; GPU stays INFO; watchdog flags fresh vs 2-hour-stale heartbeat). Off-disk mirror verified by SHA-256 in an isolation test (latest + all snapshots). Disk guard fails open. All server/OCR edits confirmed intact on the host via authoritative tools. Nothing here assumes a modern OS or GPU — lite/legacy behave identically. Run host-side VERIFY-ALL.bat to compile-check the whole tree.